Fiscal Year:	FY 2020	**Task Last Updated:**	FY 03/28/2020
PI Name:	Narayanan, Ranga Ph.D.		
Project Title:	A Novel Way to Measure Interfacial Tension Using the Electrostatic Levitation Furnace (ELF)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ranga@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-281-5382
Organization Name:	University of Florida		
PI Address 1:	Chemical Engineering, 205, PST		
PI Address 2:			
PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/22/2017	**End Date:**	05/21/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX17AL27G		
Performance Goal No.:			
Performance Goal Text:			

A novel method using an Electrostatic Levitator (EL) is proposed to determine the interfacial tension (IFT) of liquid/gas systems. This key thermophysical property is of interest for materials processing and in welding operations in space and on ground. The novel method for measurement of IFT proposed involves the use of electrostatically forced Faraday instability. In this method, an oscillatory electrostatic forcing is applied to a liquid system, resulting in sudden pattern formation at the interface, i.e., an instability is caused. The "instability" is a result of resonance between the imposed forcing frequency and the natural frequency of the system. The critical amplitude and mode shapes are predictable from linear stability theory and agree superbly with experiments. The measurement of IFT directly aligns with the goals of the Thermophysical Property Measurement emphasis of the current NRA (NASA Research Announcement).
Faraday experiments by this team using mechanical forcing have shown excellent agreement between theory and

Task Description:	experimental results. However, in the case of mechanical forcing, the force of gravity overwhelms the effects of IFT on the system stability and therefore it is almost impossible to determine the IFT with accuracy. Electrostatic forcing counters gravity, allowing the effects of IFT to play a significant role in the ultimate determination of the system stability. Accordingly, experimentally mapped stability curves can be fit accurately with theoretical curves, allowing for the determination of the IFT with great accuracy. The proposing team has successfully used electrostatic forcing in ground equipment to show that Faraday waves can be obtained as per theoretical predictions. The current state of the art uses an open container geometry to hold the liquid. However, a levitated sphere may also be used, and our proposal necessarily includes this method. We therefore propose a project with two NEW aspects. In the first stage of the research, we propose to use ground-based facilities at JAXA (Japan Aerospace Exploration Agency) and NASA Marshall Space Flight Center (MSFC) to accurately obtain the IFT and surface tension for liquids using electrostatic Faraday forcing in open containers as well as in levitated drops. In the second stage of the research, we will perform flight experiments using the Electrostatic Levitation Furnace (ELF) in the International Space Station (ISS) to obtain the threshold for instability and determine the surface tension of these systems. In summary, the research allows an alternate new method, viz., Faraday instability, for the determination of interfacial tension. Using electrostatic forcing instability will permit an independent confirmation of measurements obtained by the single-pulse oscillating-drop technique where decay rates are measured and related to interfacial tension and viscosity. Ground-based research using the electrostatic levitator available at the NASA MSFC and JAXA, and space experiments on ELF in the ISS will provide accurate data. In addition to being an alternative method for IFT measurement, the proposed technique also has versatility as a potential means in measuring interfacial tension for high viscosity materials and two-liquid interfacial tension on ground as well as in space. The proposing team has recently collaborated successfully with JAXA and has conducted experiments to study the feasibility of electrostatic Faraday instability. These results form the basis for the current proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurate interfacial tension values are crucial in processes of importance to both industry and international space missions, including: a) Additive manufacturing of metals, where IFT knowledge is pivotal to avoid shrinkage and wrinkling. Human Exploration: Technology Area (TA) TA7.2.3.5: Welding, TA12.1.2: Computationally-Designed Materials, TA12.4.2.5: Additive Manufacturing. b) Semiconductor crystal growth, where the IFT between a liquid melt and liquid encapsulant plays a large role in determining conditions where convection is present. c) Space enabling operations. One example is joining operations such as soldering, where the IFT plays an important role in the structure and stability of the weld. Accurate values for the IFT will enhance the safety and effectiveness of this type of operation in space where the microstructures formed are different than those on Earth where gravity is present. d) Enhanced oil recovery, where the IFT between injection fluid and crude oil plays an important role in the recovery rate of the oil and the prevention of viscous fingering. e) The accurate measurement of the IFT between liquids through the use of a different operational mode with minimal experimental modification can validate measurements made through the current techniques of levitation and pulse disturbance of spherical fluid masses.
Task Progress:	A self consistent benchmark to the measurement of surface tension of high temperature liquid metals has been developed with a proof-of-concept. In this method we have used the idea of resonance of levitated drops to create various patterns or modes. Each mode is correlated to surface tension. When the parametric frequency equals the natural frequency, the system undergoes an instability that is manifested by a sudden change of its configuration. The natural frequency is related to the surface tension and so the onset of the instability is used to determine the surface tension. By sweeping through a range of frequencies we can obtain different modes and each mode uniquely determines the surface tension. We can thus get a self consistent way of determining surface tension. This technique was demonstrated for a liquid levitated drop. The knowledge of this thermophysical property is extremely useful to predictive additive manufacturing.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Abstracts for Journals and Proceedings	Dinesh B, Brosius N, Zoueshtiagh F, Narayanan R. "Two-Phase Faraday Resonance with Patterned Substrates--Theory and Experiments." 2019 AIChE National Meeting, Orlando, FL, November 10-15, 2019. Abstracts, 2019 AIChE National Meeting, Orlando, FL, November 10-15, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Karpinsky Z, Matsumoto S, Ishikawa T, SanSoucie M, Narayanan R. "Interfacial Instability in Spheres By Resonance." 2019 AIChE National Meeting, Orlando, FL, November 10-15, 2019. Proceedings, 2019 AIChE National Meeting, Orlando, FL, November 10-15, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Pillai D, Narayanan R. "Long-Time Dynamics of Thin Leaky Dielectric Films: Sliding and Cascading Structures." 2019 AIChE National Meeting, Orlando, FL, November 10-15, 2019. Proceedings, 2019 AIChE National Meeting, Orlando, FL, November 10-15, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Dinesh B, Brosius N, Zoueshtiagh F, Narayanan R. "Faraday resonance over a corrugated surface: theory and experiments." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019

Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Matsumoto S, Ishikawa T, SanSoucie M, Narayanan R. "Determination of surface tension using higher-order modes of levitated liquid droplets." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Brosius N, Ward K, Narayanan R. "Faraday Instability at Liquid-Fluid Interfaces." Bifurcation and Fluid Dynamics (BIFD) Conference, Limerick, Ireland, July 16-19, 2019. BIFD Proceedings. Bifurcation and Fluid Dynamics (BIFD) Conference, Limerick, Ireland, July 16-19, 2019. , Jul-2019
Abstracts for Journals and Proceedings	Pillai D, Narayanan R. "Electrostatic forcing of thin liquid films." Bifurcation and Fluid Dynamics (BIFD) Conference, Limerick, Ireland, July 16-19, 2019 BIFD Proceedings. Bifurcation and Fluid Dynamics (BIFD) Conference, Limerick, Ireland, July 16-19, 2019. , Jul-2019
Abstracts for Journals and Proceedings	Pillai D, Picardo J, Narayanan R. "A reduced-order model for thermoviscous fingering of magma flow in volcanic fissures. " Bifurcation and Fluid Dynamics (BIFD) Conference, Limerick, Ireland, July 16-19, 2019. BIFD Proceedings. Bifurcation and Fluid Dynamics (BIFD) Conference, Limerick, Ireland, July 16-19, 2019. , Jul-2019
Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Matsumoto S, Ishikawa T, SanSoucie M, Narayanan R. "The Faraday forcing method of obtaining resonant frequencies in electrostatically levitated droplets." 2019 International Space Station Research and Development Conference (ISSRDC), Atlanta, GA, July 29-August 1, 2019. ISSRDC Proceedings. 2019 International Space Station Research and Development Conference (ISSRDC), Atlanta, GA, July 29-August 1, 2019. , Jul-2019
Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Karpinsky Z, Matsumoto S, Ishikawa T, SanSoucie M, Narayanan R. "Interfacial instability in spheres by resonance." Presentation at 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. Bulletin of the American Physical Society. 2019 Nov;64(13):Q37.00005. http://meetings.aps.org/Meeting/DFD19/Session/Q37.5 , Nov-2019
Articles in Peer-reviewed Journals	Pillai DS, Narayanan R. "Electrostatic forcing of thin leaky dielectric films under periodic and steady fields." J Fluid Mech. 2020 May 10;890:A20. https://doi.org/10.1017/jfm.2020.112 , May-2020
Articles in Peer-reviewed Journals	Ward KL, Zoueshtiagh F, Narayanan R. "The Faraday instability in rectangular and annular geometries: comparison of experiments with theory." Experiments in Fluids. 2019 Mar;60:53. https://doi.org/10.1007/s00348-019-2695-4 , Mar-2019
Articles in Peer-reviewed Journals	Ward KL, Zoueshtiagh F, Narayanan R. "Faraday instability in double-interface fluid layers." Physical Review Fluids. 2019 Aprl4(4):043903. https://doi.org/10.1103/PhysRevFluids.4.043903 , Apr-2019